

Nordic ID Sampo S2

Nordic ID Sampo S2 is a versatile fixed UHF RFID reader with an integrated antenna featuring software-controllable gain. Supporting both EU and US frequency bands in a single device, it is suitable for POS, gates, and various integration use cases with its embedded Linux computer.



UHF RFID

Supported standard	ISO 18000-63 (EPC Class 1 Gen2 V2), AES authentication ISO/IEC 29167-10
Frequency band	ETSI 865.6–867.6 MHz and FCC/IC 902–928 MHz
Regulatory	CE ETSI EN 302 208, CE ETSI EN 301 489, FCC part 15.247 IC RSS-210, Safety IEC 62368-1:2019 ed. 3
Typical reading speed	200 tags/s in DRM mode
Radiated power (integrated antenna)	Max 30 dBm (1 W) ERP / 1.6 W EIRP; Min 1 dBm ERP
Integrated antenna features	Wide band ETSI + FCC/IC, 7 dBic circular polarized, Beam width 80°, Nominal reading distance 5 m
Conducted power for external antenna ports	27 dBm (500 mW)
External antenna port	3 pcs. 50 Ω / SMA Female
Isolation between external antenna ports	35 dB / typical

PLATFORM

CPU	1.2GHz 64-bit Quad-Core Cortex-A53 processor
Operating system	Debian based embedded Linux
Memory	1 GB RAM / 4 GB flash

USER INTERFACE

USB	USB host, type A / USB device, type B, USB HID class supported
SIM	Mini-SIM
Indicators	4 pcs (programmable) LEDs; Buzzer; Capacitive sensor for triggering

CONNECTIVITY

GPIO	4 inputs / 4 outputs, opto isolated
Wireless WAN (optional)	3G HSPA / 2G GPRS / EDGE
LAN	Ethernet 10/100 Mbit
Wireless LAN (optional)	IEEE 802.11 a/b/g/n, dual band 2.4 / 5 GHz

POWER

External Power Supply	PoE 802.3af or AC/DC adapter: 100-240 VAC / 24 VDC, 1.25A
Operating Power	12W PoE, 20 W DC

SIZE AND WEIGHT

Dimensions	(W) 200 x (L) 260 x (H) 25 mm
Weight	530 g / 18.7 oz

ENVIRONMENT

Environmental sealing	IP20, for indoor use only
Operating Temperature	-20 to 55 °C
Storage Temperature	-40 to 85 °C

SOFTWARE INTERFACE

Data management	Nordic ID Radea
Device management	Nordic ID Radea, SOTI®
API support	NUR API for RFID and RESTful service
Other protocols	LLRP

FEATURE HIGHLIGHTS

- Effective fixed UHF RFID reader for a multitude of use cases
- SW controllable tx power for low and normal gain functionalities
- Integrated Linux based computer for versatile SW development
- One device for usage in EU or US frequency environments
- Versatile host and device connectivity options
- Avoid cross readings due to high isolation between antenna ports

PRODUCT GROUP

Category	RFID Reader
Product Group	Fixed UHF RFID Reader